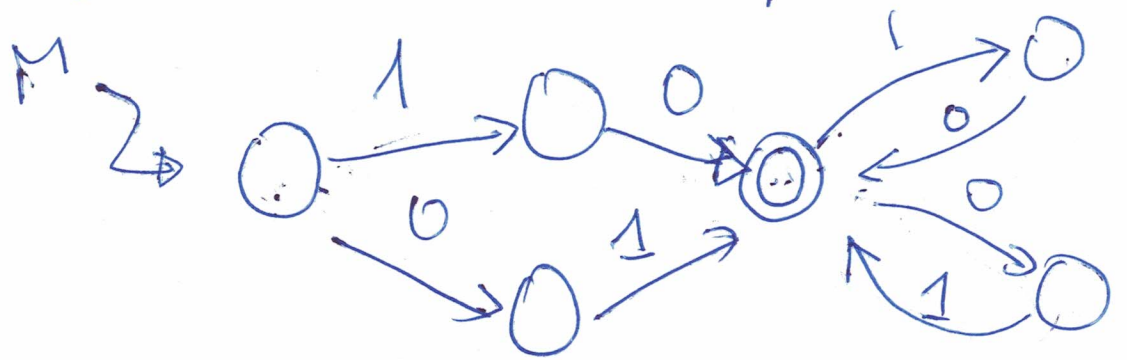
Cpts 317. 9/16

Examples on how to write regExp from a FA.

Given: a regExp $(10+01)(10+01)^*$

Want: a FA to accept it.



Given: a FA shown above.

Want: a regExp for L(M).

$$L(M) = (10 + 01) \cdot (10 + 01)^* .$$

Wrong answer: $L(M) = (10 + 01) \cdot ((10)^* + (01)^*)$

check: $100110 \in L(M)$,

but $100110 \notin$ the regExp

Descriptions of FA:

1. in English

2. in diagram + English.

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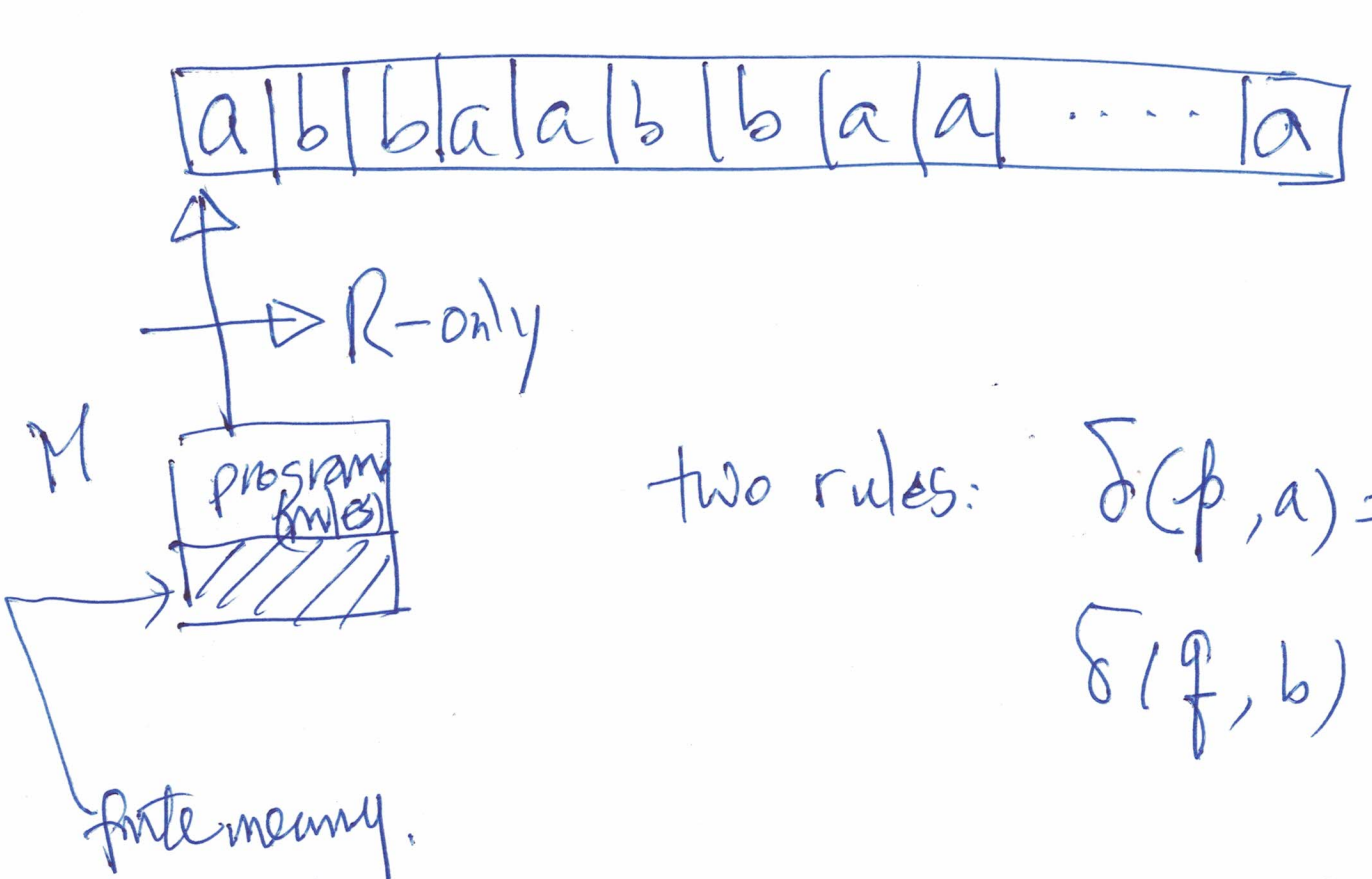
graph.



input tape with input word.

two rules: $\delta(p, a) = q$.

$\delta(q, b) = p$.

memory size = 1 bit

$= \log_2 2$ bit.

Cartesian Product Construction.

Thm. Let M_1 and M_2 be two DFAs. Then, we can construct a DFA M s.t.

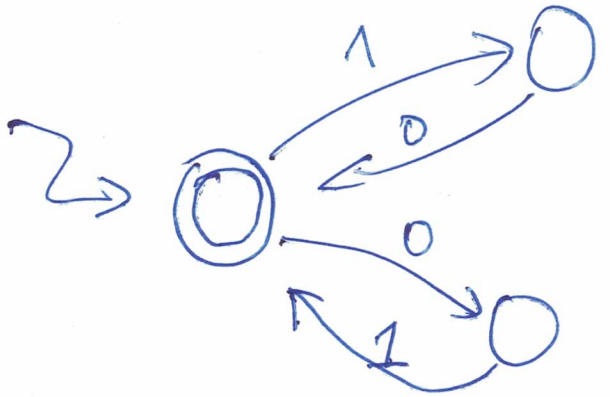
$$L(M) = L(M_1) \cap L(M_2).$$

Example. $L_1 = (10 + 01)^*$, $L_2 = (11 + 0)^*$.

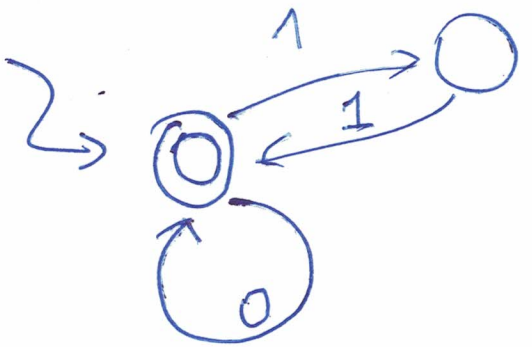
I want to know:

$$L = L_1 \cap L_2 = ?$$

DFA M_1 for L_1:



DFA M_2 for L_2:

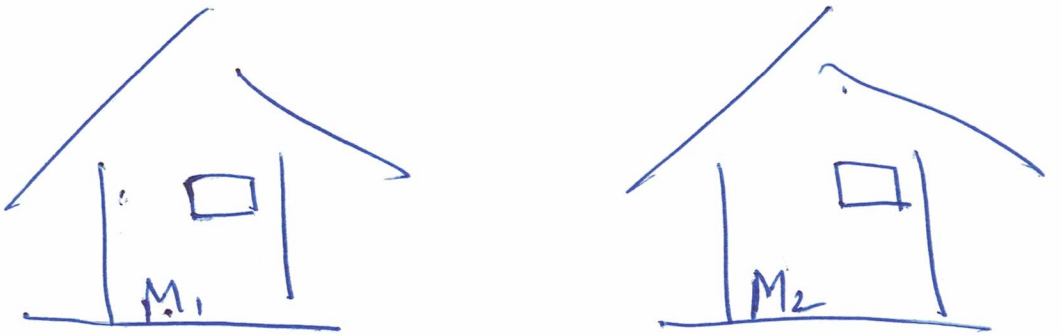


From the theorem, we can construct a DFA M

M

s.t. $L(M) = L_1 \cap L_2$.

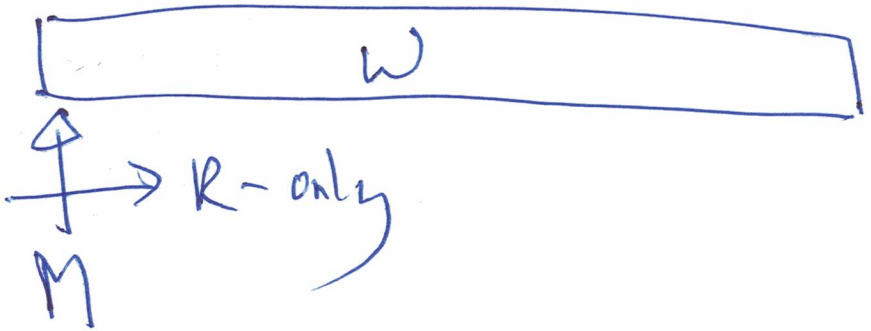
Given:



Want to construct M s.t.

$$L(M) = L(M_1) \cap L(M_2).$$

Proof (of the theorem). (rigorous proof in English). We will construct the M as follows. On an input w, M runs



M_1 and M_2 in parallel, while, for each symbol that M reads, M feeds the symbol to M_1 and to M_2, resp. At the

end of input w, if both M_1 and M_2 say yes, then M says yes. Clearly, M accepts exactly $L_1 \cap L_2$. $\blacksquare$

Challenge of today:

From the previous construction of M from M_1 & M_2,

Suppose that M_1 uses 2 bits of mem,

M_2 uses 4 bits of mem.

Then, M uses ? bits ?

$\parallel$

6.

Key: state # in M $= 2^6 = 2^2 \times 2^4$

$=$ state # in M_1 $\times$ state # in M_2.